Final Design Project – Guidelines and Instructions

1. Project overview

The final project is your opportunity to apply the knowledge and practical skills you have acquired throughout the course. You will design, implement, and demonstrate a functional embedded system that integrates hardware and software components to solve a practical problem or perform a specific task. You will complete the project through several milestones, which give you the chance to receive feedback, demonstrate progress, and improve your design before the final evaluation.

For inspiration, https://people.ece.cornell.edu/land/courses/ece4760/FinalProjects/. Projects are not expected to reach the same level of complexity or sophistication; choose a feasible scope appropriate to this course.

2. Requirements

1. Implement a physical prototype using a microcontroller and relevant electronic components.
2. Develop the project software using either C or Assembly language. Students may choose either language or must be able to explain their code.
3. Integrate the hardware and software into a working system.
4. Test the system and document verification results.
5. Present a working prototype in the final live demonstration.

Project-specific requirements will be finalized during proposal approval. A simulation-only project requires prior instructor approval.

3. Timeline and Milestones

Milestone 1 – Project Proposal

Each team must submit a proposal and obtain approval before proceeding with full implementation or significant purchases.

- Project title and team members (names and student IDs)
- Problem statement, objectives, and expected functions
- Preliminary system block diagram and implementation approach
- Component list, estimated budget, and proposed suppliers

Deadline is 5th of Nov.

Milestone 2 – Check-off 1

By this check-off, each team must have ordered or purchased the components needed for the project. Submit a short report (about one page) covering:

- Order or purchase all project components.
- Show completed work and current functionality.

- Explain problems encountered and troubleshooting performed.
- Maintain a brief log of progress and individual contributions.

Milestone 3 – Check-off 2

Submit a short report (about one page) covering:

- Show completed work and current functionality.
- Explain problems encountered and troubleshooting performed.
- Identify remaining tasks and next-week objectives.
- Maintain a brief log of progress and individual contributions.

Milestone 4 – Final Report and Live Demonstration

Final live demonstration:

- Present a fully integrated prototype and demonstrate its main functions.
- Explain system architecture, circuit connections, and program operation.
- Show relevant testing results and discuss limitations.
- Answer technical questions and explain individual contributions.

Final report: Submit a final report with introduction, objectives, system block diagram, implementation, tests/results, limitations, conclusion, and references. Include the complete source code (C or assembly) and relevant design files.

4. Purchasing and Local Sourcing

Shipping delays are the most common reason projects slip. Please source parts locally and order early.

- **iElectrony:** [https://ielectrony.com/en]
- **Amazon SA**
- **iHub / DRL:** [https://drl.dtv.sa/services]
- **Emdaad (EE store):** [https://emdaad.kfupm.edu.sa/en/student/catalog/]

5. Budget and Reimbursement

Subject to departmental confirmation, reimbursement of up to 200 SAR per team may be available for eligible project expenses after a successful working final demonstration. Students should retain original invoices and receipts. Eligibility, documentation, and approval procedures will be announced if the policy is confirmed. Reimbursement is not guaranteed.

6. Optional PCB Design Bonus (Week 13/14)

Teams that want extra credit may turn part of their prototype into a custom PCB.

- **Deliverables:** schematic, board layout, and exported manufacturing files (e.g., Gerber), [by the end of Week 13/14].

- **What we look for:** a correct schematic, sensible component placement and routing, passing design-rule checks, and a short note on design decisions.
- **Fabrication:** boards are fabricated by the Dream Realization Lab (DRL). Submit your fabrication request through the DRL services page (https://drl.dtv.sa/services).
- **PCB design software:** you may use PCB design software such as KiCad to create your schematic, board layout, and manufacturing files.
- **Bonus value:** [+X points].

8. AI Use Policy

You may use AI tools in this project, as long as you are transparent about it and you understand everything you submit. You are responsible for all code and hardware in your project, and the staff may ask you to explain any part of it.

9. Evaluation Criteria

Criterion	Description
Project proposal	Feasibility, clarity, completeness
Check-off 1, 2	Demonstrated work and consistent development
Hardware implementation	Correctness and build quality
Software implementation	Functionality and code quality
System integration	Successful hardware–software interaction
Final demonstration	Reliability and technical understanding
Final report	Documentation and testing evidence
Individual contribution	Participation and understanding